

CSE 4621

Machine Learning

Topic: Basic Practices in ML

Source: [DeepLearning.AI](https://deeplearning.ai)

Presented by

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Training set, validation set, test set

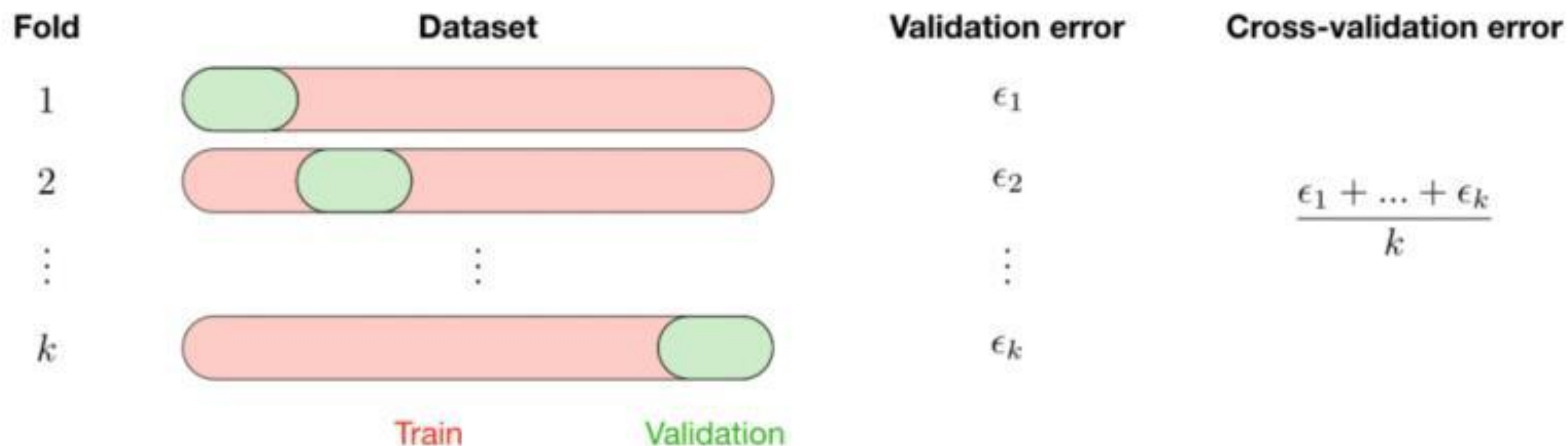
Training set	Validation set	Testing set
- Model is trained - Usually 80% of the dataset	- Model is assessed - Usually 20% of the dataset - Also called hold-out or development set	- Model gives predictions - Unseen data

Once the model has been chosen, it is trained on the entire dataset and tested on the unseen test set. These are represented in the figure below:



Cross-validation

- Holdout Validation
- LOOCV (Leave One Out Cross Validation)
- K-Fold Cross Validation
- Stratified Cross-Validation



Hyperparameter Tuning

Hyperparameters are defined as the parameters that are explicitly defined by the user to control the learning process.

- These are external to the model, and their values cannot be changed during the training process.
- Few examples:
 - The k in kNN or K-Nearest Neighbour algorithm
 - Learning rate
 - Train-test split ratio
 - Batch Size
 - Number of Epochs
 - Branches in Decision Tree
 - Number of clusters in Clustering Algorithm

Model Performance Assessment

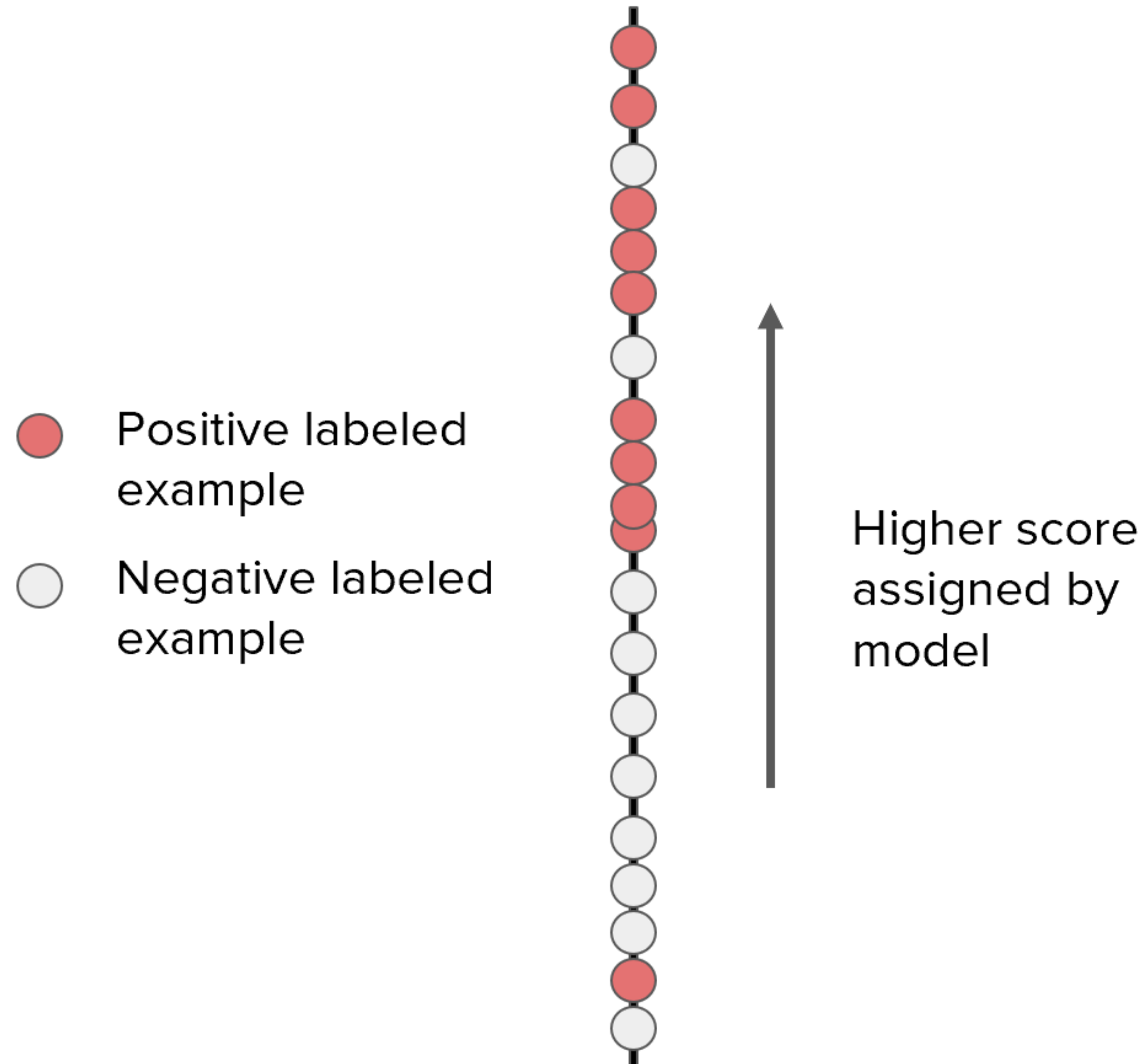
Metric	Formula	Interpretation
Accuracy	$\frac{TP + TN}{TP + TN + FP + FN}$	Overall performance of model
Precision	$\frac{TP}{TP + FP}$	How accurate the positive predictions are
Recall Sensitivity	$\frac{TP}{TP + FN}$	Coverage of actual positive sample
Specificity	$\frac{TN}{TN + FP}$	Coverage of actual negative sample
F1 score	$\frac{2TP}{2TP + FP + FN}$	Hybrid metric useful for unbalanced classes

True Class		Positive	Negative
Predicted Class	Positive	TP	FP
	Negative	FN	TN

Why should we care about evaluation metrics?

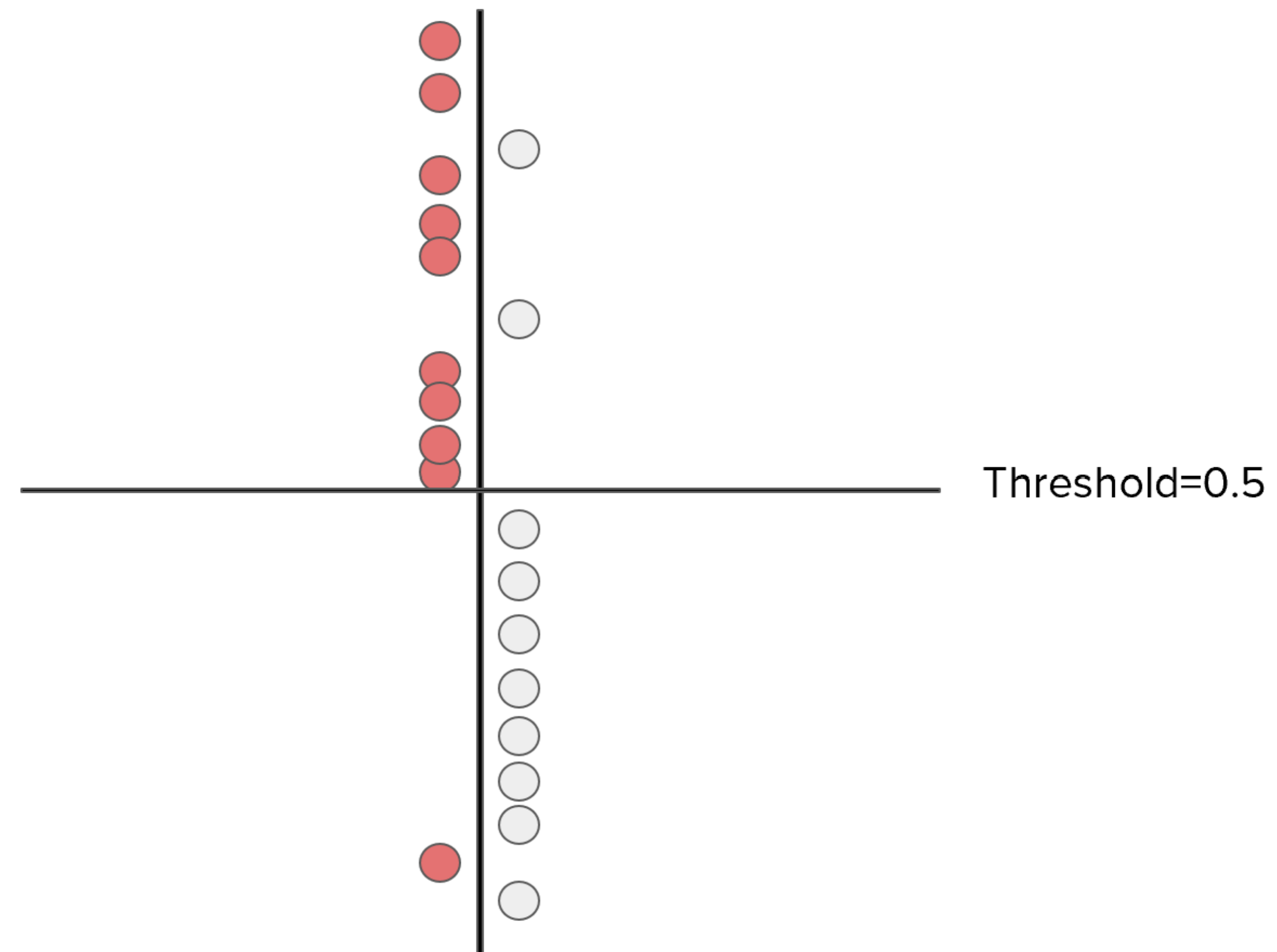
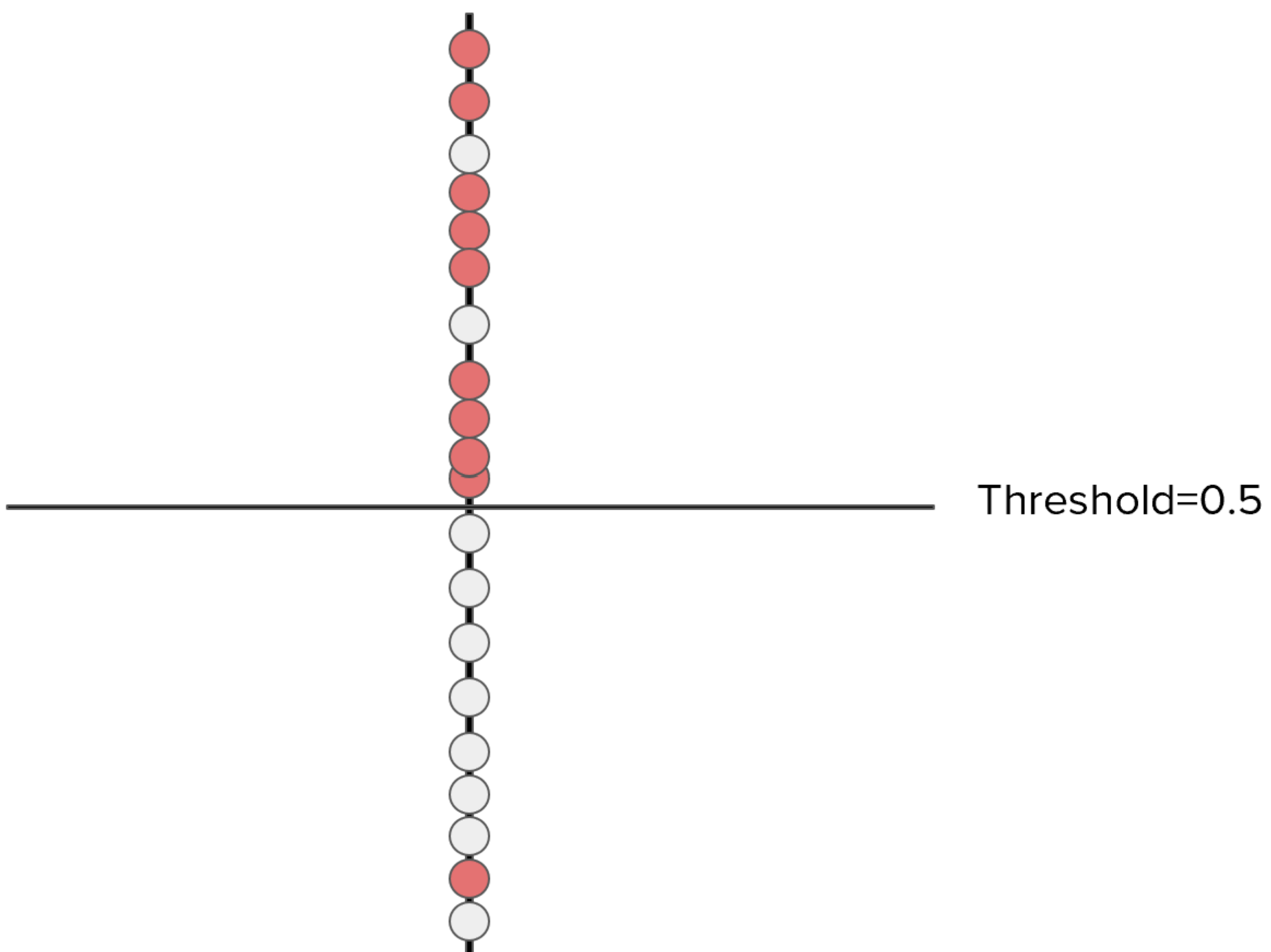
- Quantitatively define a **real world objective**
 - Training objective (loss function) is typically only a *proxy* for this objective
 - Ideally this objective matches the real world objective as closely as possible
- Help organize ML team effort toward that target
 - Trying to improve the metric on the *validation set*
- Quantify the gap between
 - Baseline and desired performance (initial difficulty estimate)
 - Current performance and desired performance
- Help debug

Score-based Models (Classifier)

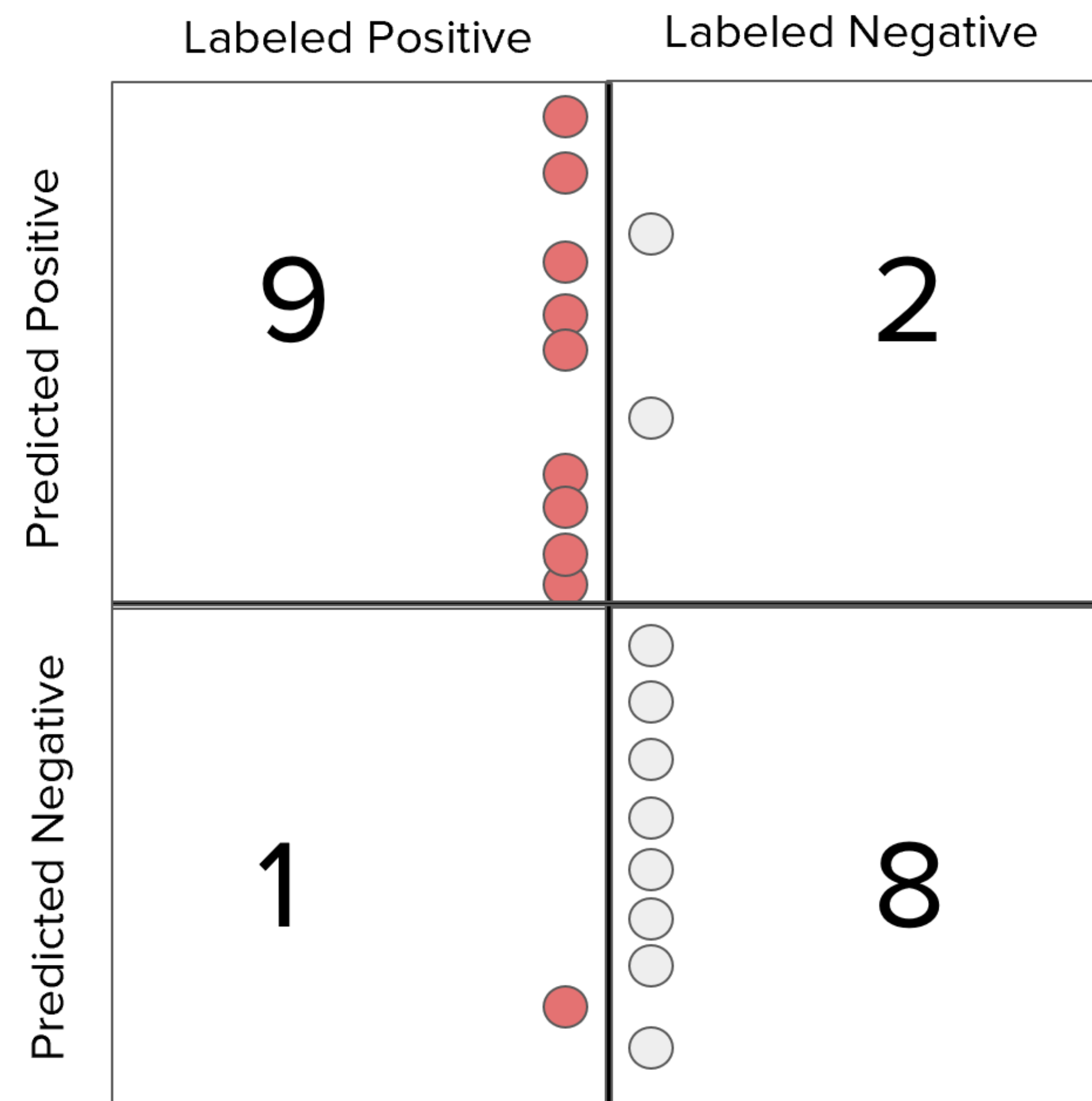
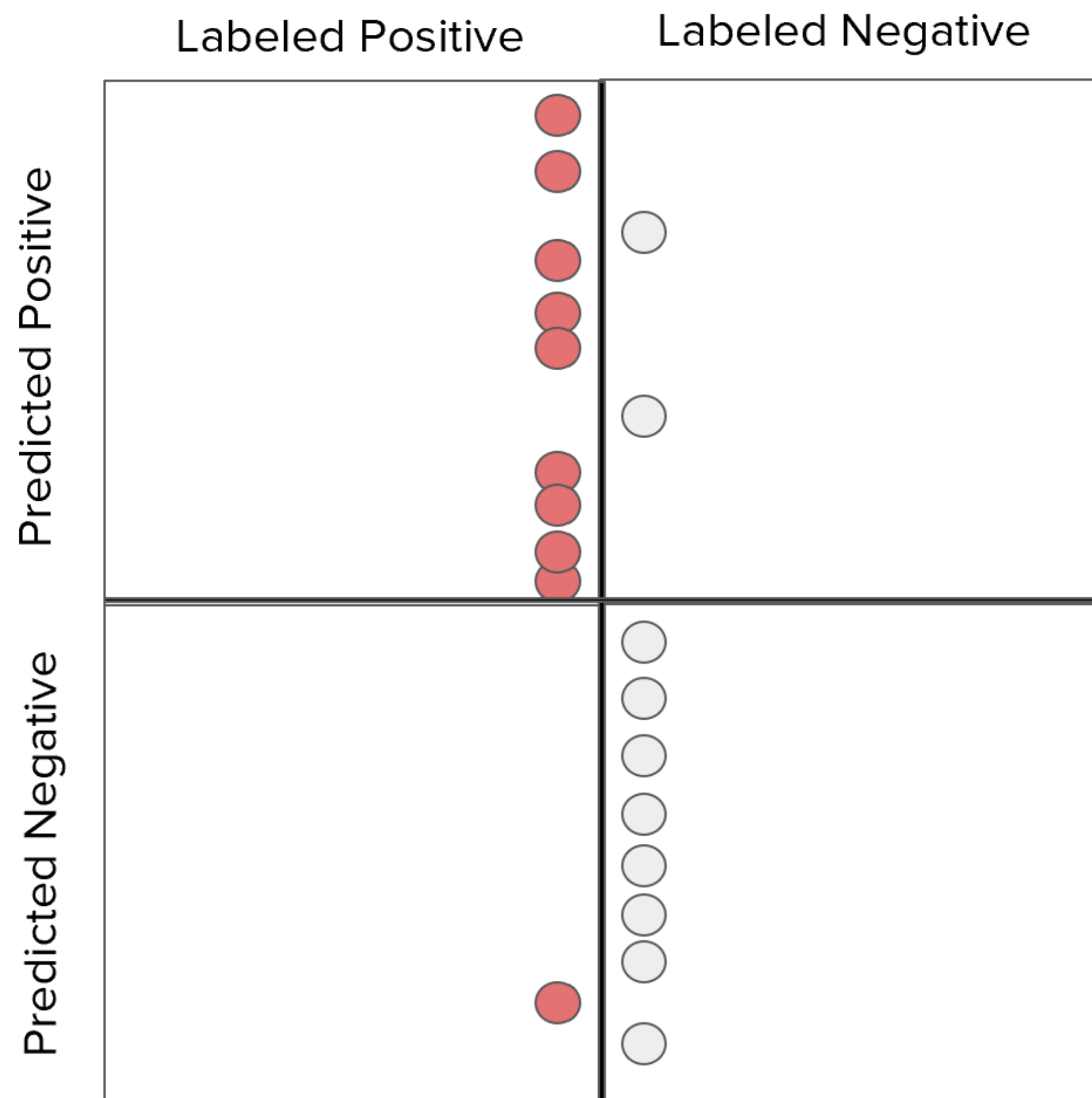


- Score output by a logistic regression model or SVM
- Need to pick a threshold to have a classifier

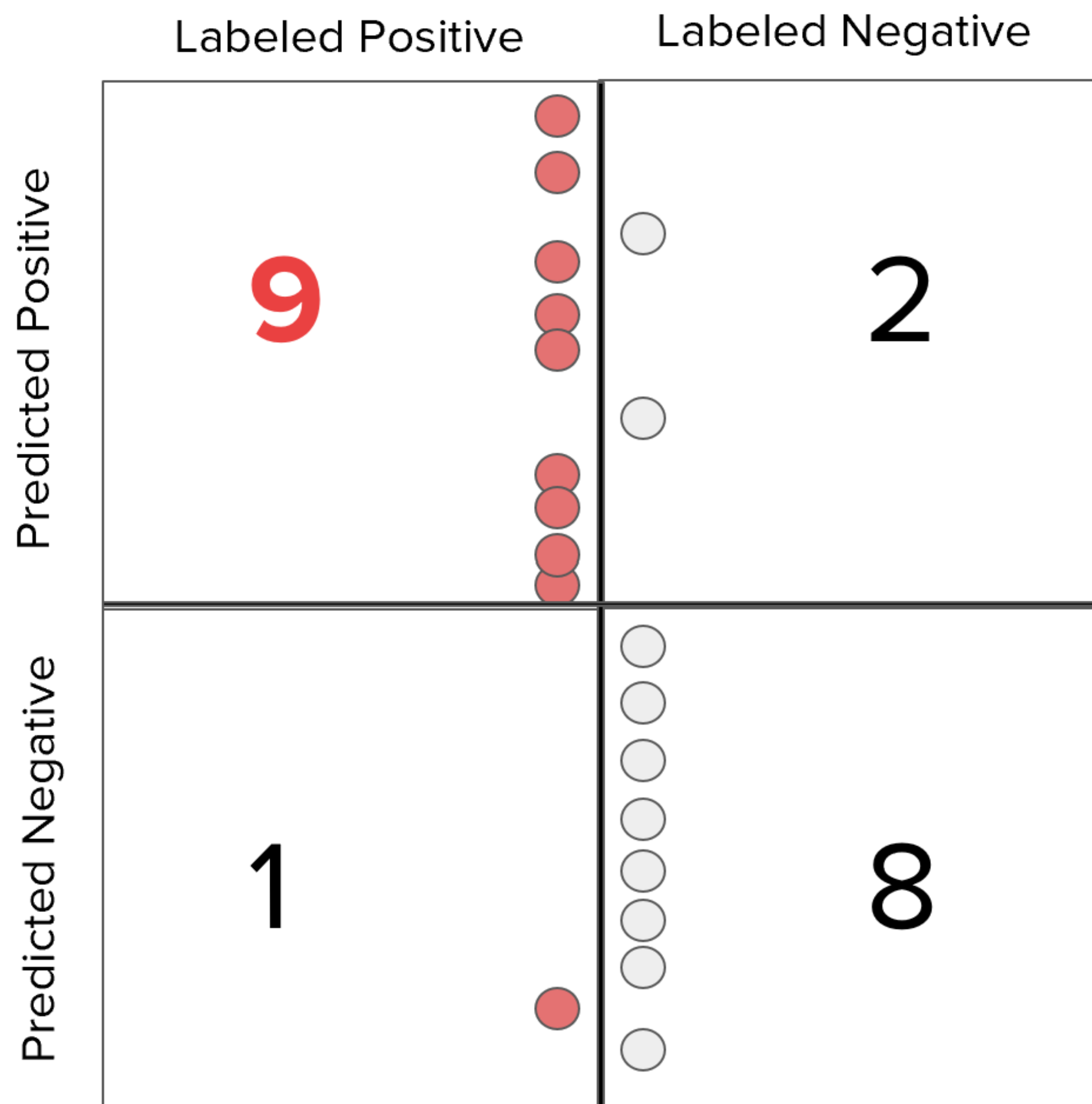
Score-based Models (Classifier)



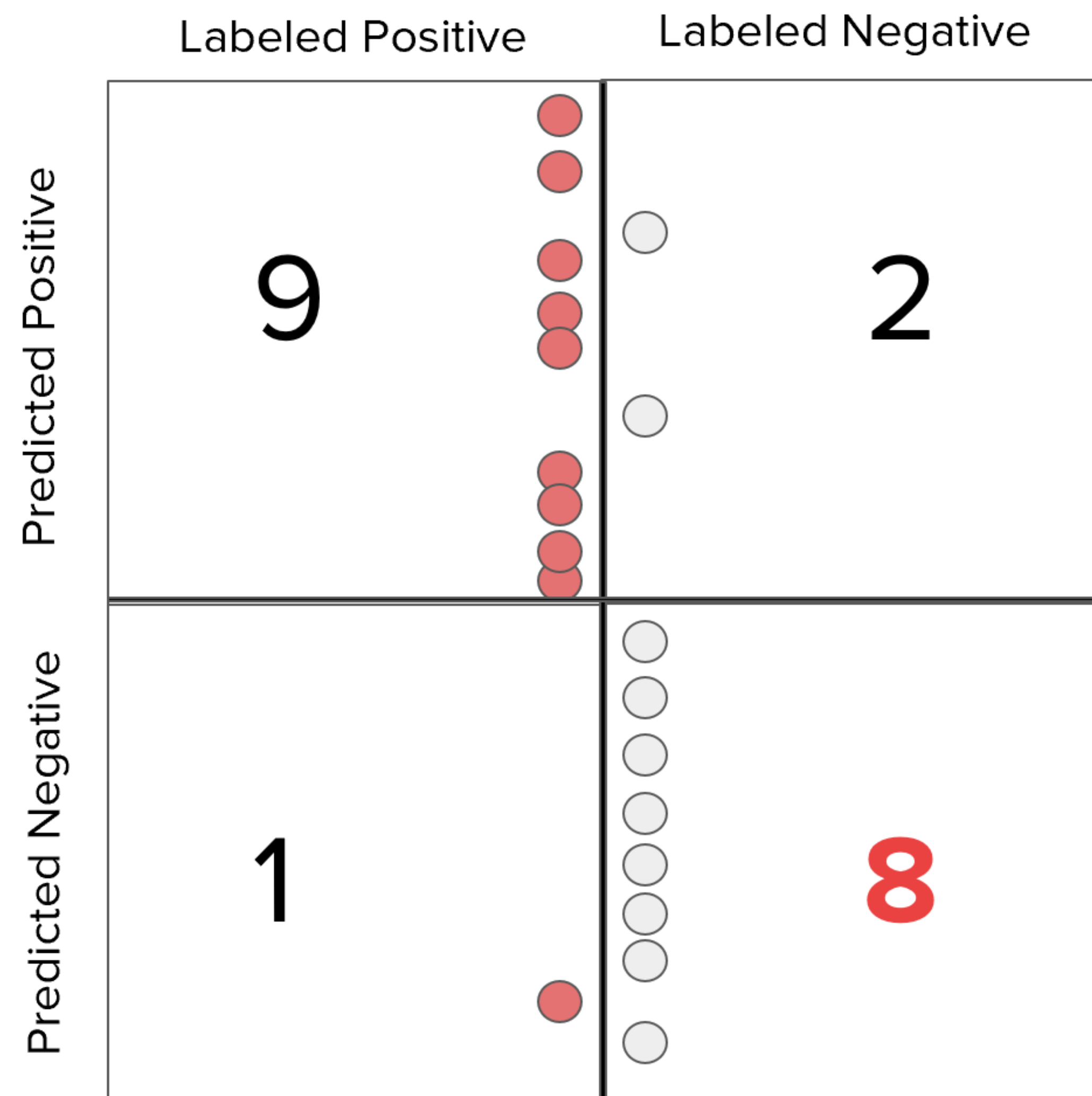
Confusion Matrix



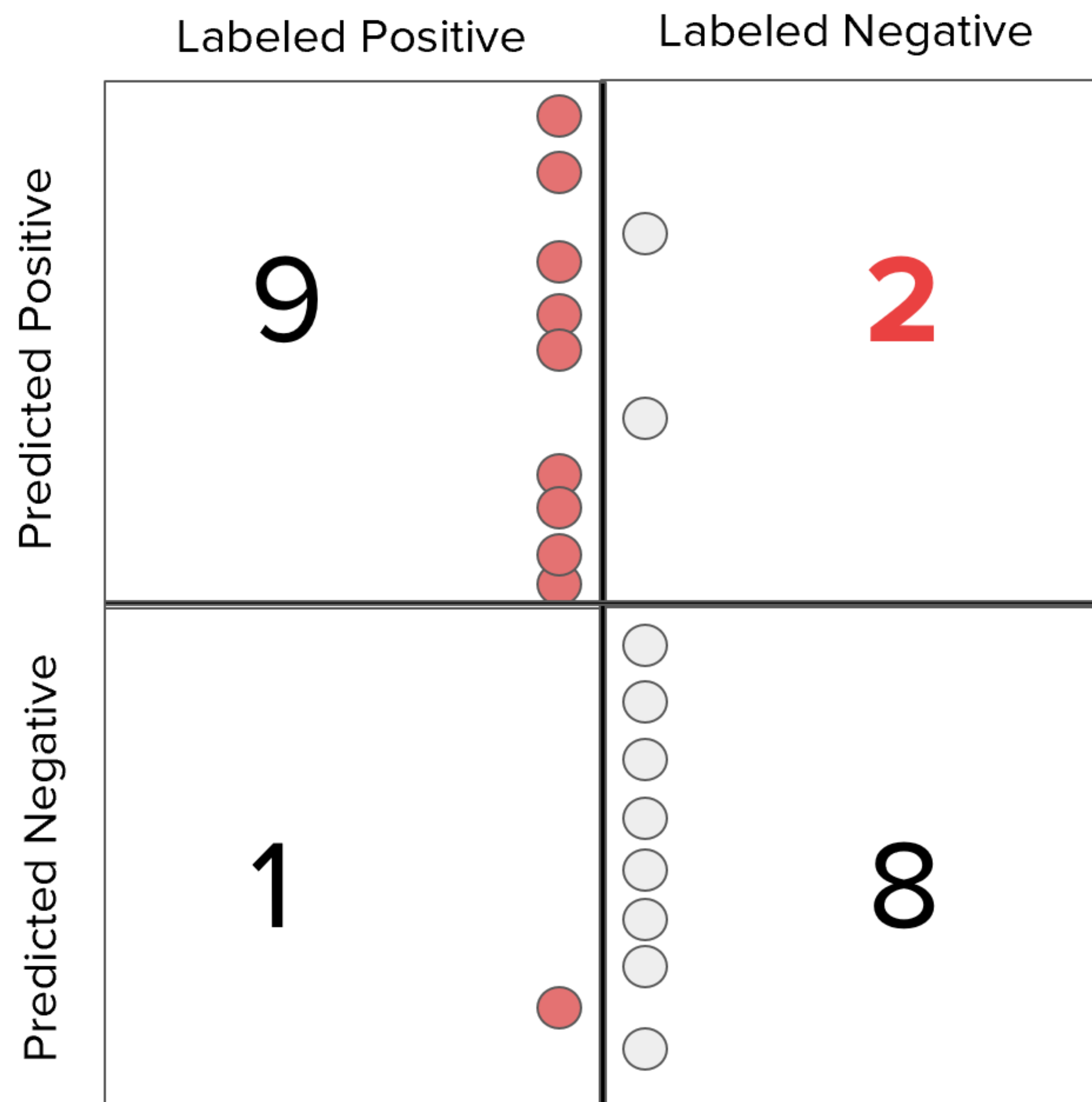
Point Metrics: True Positives



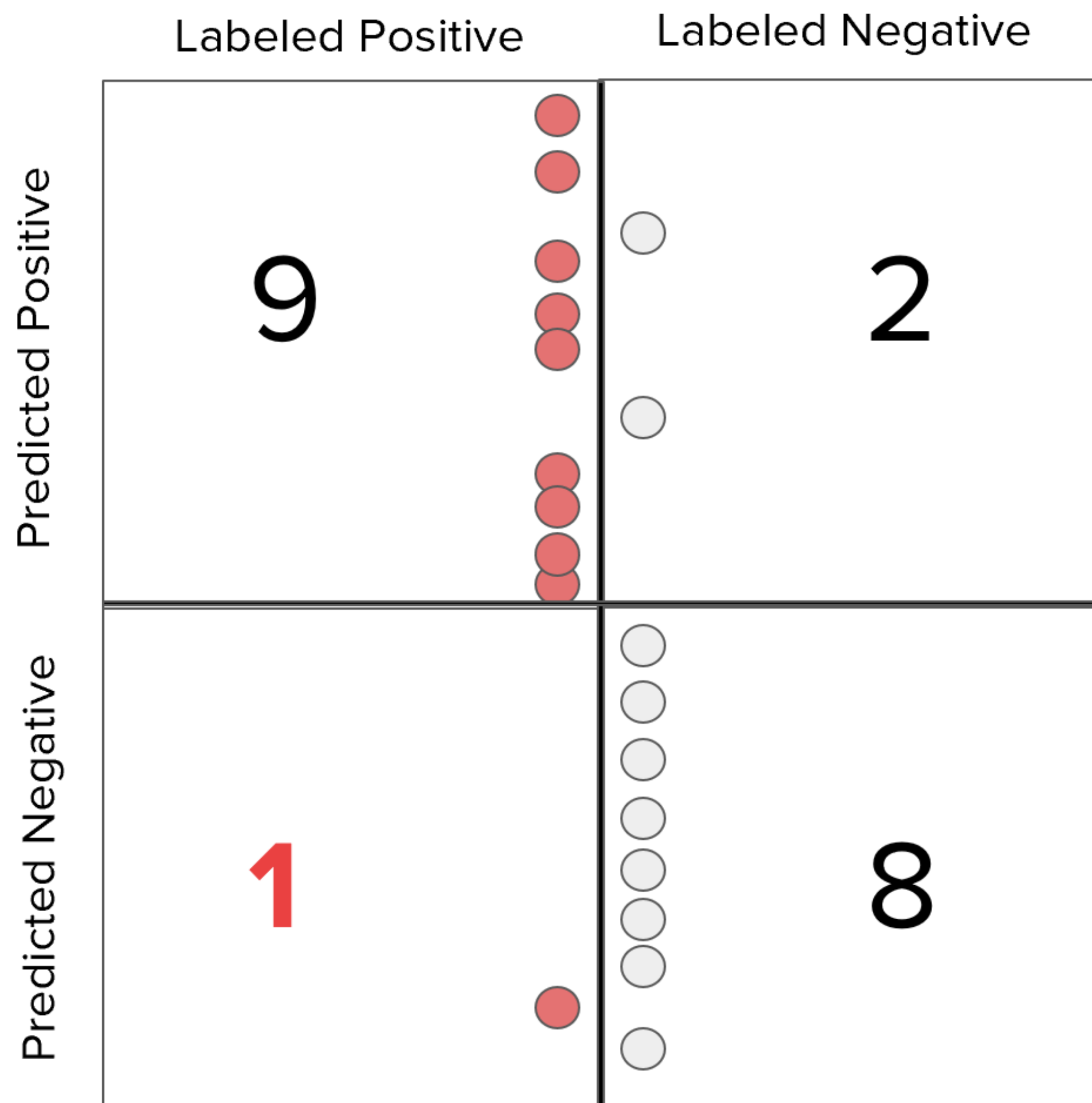
Point Metrics: True Negatives



Point Metrics: False Positives



Point Metrics: False Negatives



Point Metrics: Accuracy

	Labeled Positive	Labeled Negative
Predicted Positive	9	2
Predicted Negative	1	8

$$\begin{aligned}\text{Accuracy} &= (\text{TP} + \text{TN}) / \text{Total} \\ &= (9 + 8) / 20 \\ &= 0.85\end{aligned}$$

Point Metrics: Precision

	Labeled Positive	Labeled Negative
Predicted Positive	 9	 2
Predicted Negative	 1	 8

$$\begin{aligned}
 \text{Precision} &= \text{TP} / (\text{TP} + \text{FP}) \\
 &= 9 / (9 + 2) \\
 &= 0.82
 \end{aligned}$$

Also called PPV (Positive Predictive Value)

Point Metrics: Positive Recall

	Labeled Positive	Labeled Negative
Predicted Positive	 9	 2
Predicted Negative	 1	 8

$$\begin{aligned}
 \text{Recall} &= \text{TP} / (\text{TP} + \text{FN}) \\
 &= 9 / (9 + 1) \\
 &= 0.9
 \end{aligned}$$

Also called sensitivity

Point Metrics: Negative Recall

	Labeled Positive	Labeled Negative
Predicted Positive	9 	2 
Predicted Negative	1 	8 

$$\begin{aligned}
 \text{Negative Recall} &= \text{TN} / (\text{TN} + \text{FP}) \\
 &= 8 / (8 + 2) \\
 &= 0.8
 \end{aligned}$$

Also called specificity

Point Metrics: F-score

Summarize precision and recall into a single score

F1-score: Harmonic mean of precision and recall

$$\text{precision} = \frac{TP}{TP + FP}$$

$$\text{recall} = \frac{TP}{TP + FN}$$

$$F_1 = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$$

Point Metrics: F-score

Example	Actual		Total	
Model Prediction	80	5	85	Precision = (80/85) = 0.94
	10	5	15	
Total	90	10	100	
	Recall = (80/90) = 0.88		Accuracy = (80+5)/100 = 85%	F1 = 0.91

Point Metrics: F-score

Example 1: Consider the below case where there are only 9 cases of true positives out of a dataset of 100.

	Actual		Total
Model Prediction	1	1	2
	8	90	98
Total	9	91	100

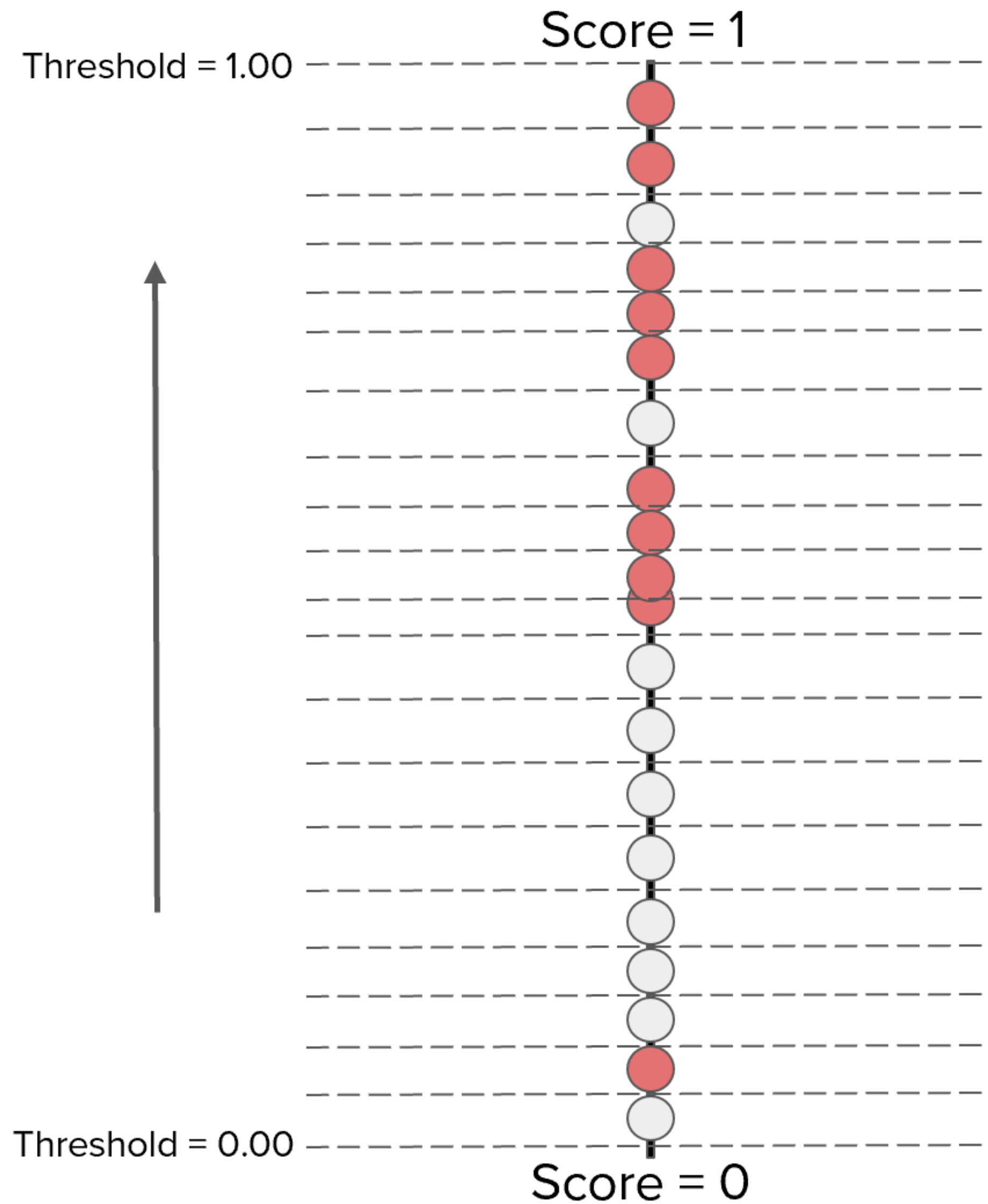
Point Metrics: F-score

Example 2: However one must also consider the opposite case where the positives outweigh the negative cases. In such a case our model will try to predict everything as positive.

	Actual		Total
Model Prediction	90	8	98
	1	1	2
Total	91	9	100



Point Metrics: Varying the threshold



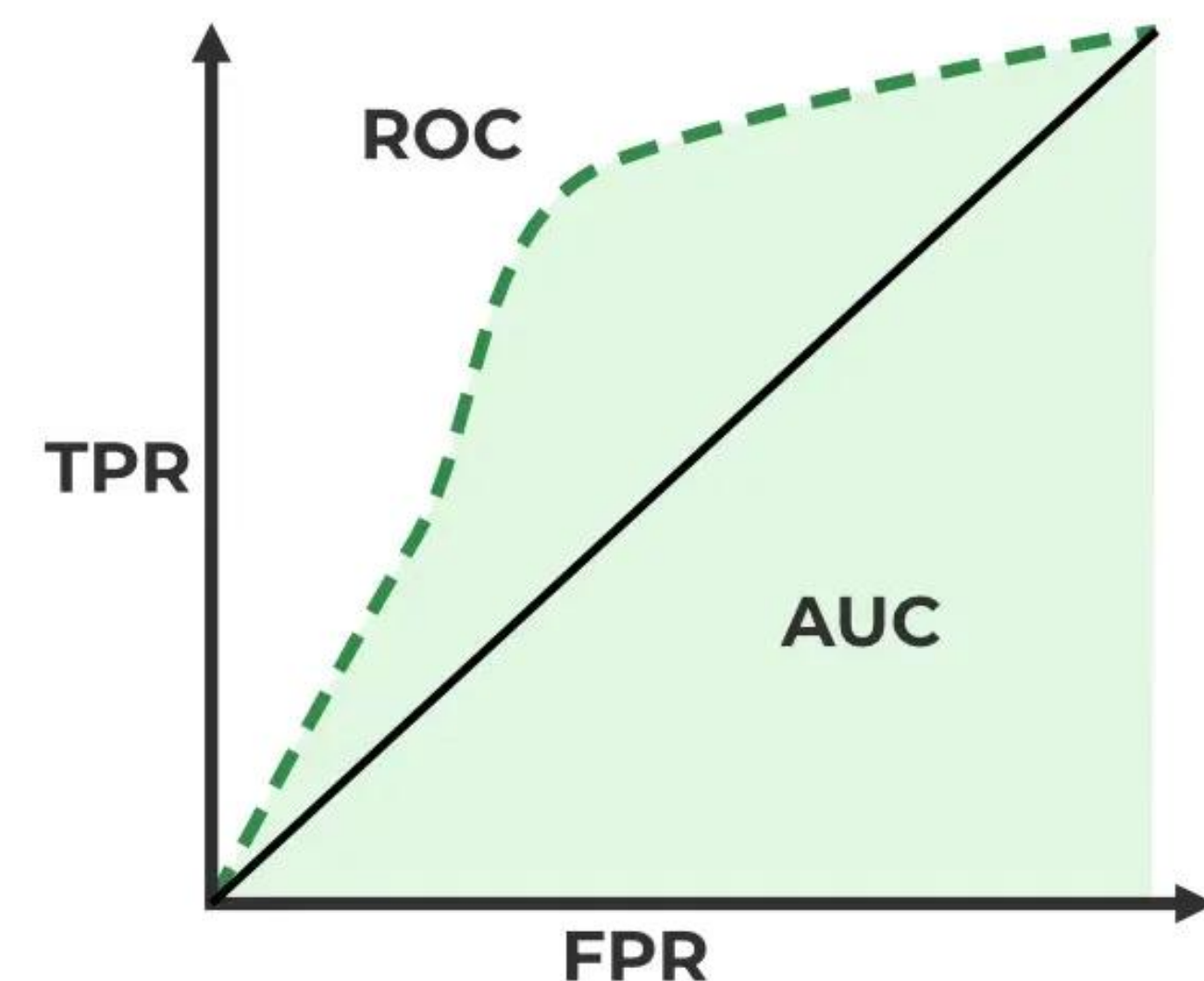
Threshold	TP	TN	FP	FN	Accuracy	Precision	Recall	Specificity	F1
1.00	0	10	0	10	0.50	1	0	1	0
0.95	1	10	0	9	0.55	1	0.1	1	0.182
0.90	2	10	0	8	0.60	1	0.2	1	0.333
0.85	2	9	1	8	0.55	0.667	0.2	0.9	0.308
0.80	3	9	1	7	0.60	0.750	0.3	0.9	0.429
0.75	4	9	1	6	0.65	0.800	0.4	0.9	0.533
0.70	5	9	1	5	0.70	0.833	0.5	0.9	0.625
0.65	5	8	2	5	0.65	0.714	0.5	0.8	0.588
0.60	6	8	2	4	0.70	0.750	0.6	0.8	0.667
0.55	7	8	2	3	0.75	0.778	0.7	0.8	0.737
0.50	8	8	2	2	0.80	0.800	0.8	0.8	0.800
0.45	9	8	2	1	0.85	0.818	0.9	0.8	0.857
0.40	9	7	3	1	0.80	0.750	0.9	0.7	0.818
0.35	9	6	4	1	0.75	0.692	0.9	0.6	0.783
0.30	9	5	5	1	0.70	0.643	0.9	0.5	0.750
0.25	9	4	6	1	0.65	0.600	0.9	0.4	0.720
0.20	9	3	7	1	0.60	0.562	0.9	0.3	0.692
0.15	9	2	8	1	0.55	0.529	0.9	0.2	0.667
0.10	9	1	9	1	0.50	0.500	0.9	0.1	0.643
0.05	10	1	9	0	0.55	0.526	1	0.1	0.690
0.00	10	0	10	0	0.50	0.500	1	0	0.667

ROC AUC

ROC: Receiver Operating Characteristic curve (ROC). It plots TPR vs. FPR at different thresholds. It represents the trade-off between the sensitivity and specificity of a classifier.

AUC: Area under the ROC curve (AUC). A higher AUC value indicates better model performance as it suggests a greater ability to distinguish between classes. An AUC value of 1.0 indicates perfect performance while 0.5 suggests it is random guessing.

Metric	Formula	Equivalent
True Positive Rate TPR	$\frac{TP}{TP + FN}$	Recall, sensitivity
False Positive Rate FPR	$\frac{FP}{TN + FP}$	1-specificity



Class Imbalance: Problems

Symptom: prevalence $< 5\%$

Metrics lose meaning

Inhibits learning

e.g. logistic regression can be overwhelmed by majority class



Class Imbalance: Problems

Example:

- Predict if a bowler will not bowl a no-ball?
- Assuming 15 no-balls in an inning, a model that says 'Yes' all the time will have 95% accuracy.
- Total points: 315 (assuming other balls are legal 😊)
- No-ball label: Class 0 (4.76% are from this class)
- Not a no-ball label: Class 1 (95.24% are from this class)
- Using accuracy as performance metric, we can say that a model is very accurate, but it is not useful or valuable in fact.

Matthew's Correlation Coefficient (MCC)

Matthew's correlation coefficient determines the correlation between true class and predicted class. The higher the correlation between true and predicted values, the better the prediction.

$$\text{MCC} = \frac{(\text{TP})(\text{TN}) - (\text{FP})(\text{FN})}{\sqrt{(\text{TP} + \text{FN})(\text{TP} + \text{FP})(\text{TN} + \text{FN})(\text{TN} + \text{FP})}}, \quad |\text{MCC}| \leq 1$$

	Actual		Total
Model Prediction	90	8	98
	1	1	2
Total	91	9	100

Multiclass Classification

Predict if a bowler will bowl a no-ball, wide bowl, regular bowl?

15 no-balls, 20 wide-balls in an inning (Total balls: 335)

– Model Predictions:

		<i>Actual</i>			<i>Precision</i>
		No-ball	Wide-ball	Regular ball	
<i>Classifier Output</i>	No-ball	8	5	20	$\frac{8}{8+5+20}$
	Wide-ball	2	10	10	$\frac{10}{2+10+10}$
	Regular ball	5	5	270	$\frac{270}{5+5+270}$
<i>Recall</i>		$\frac{8}{8+2+5}$	$\frac{10}{5+10+5}$	$\frac{270}{20+10+270}$	

Thank You

Any questions?

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